

EC 551 Advanced Digital Design with Verilog and FPGAs

Fall 2026, Prof. Douglas Densmore

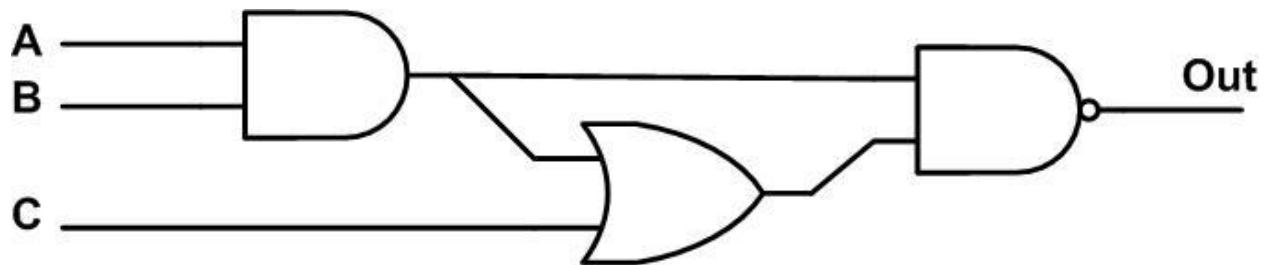
Combinational Logic Take Home Quiz

1. Convert 535_{10} to binary

2. 10111010_2 (unsigned) to decimal

3. Represent -745_{10} a Two's Complement binary number

4. Convert the following circuit to a Boolean algebraic expression by examination.



5. Create a truth table representation for the circuit.
6. Express the circuit as a canonical SOP Boolean algebraic expression.
7. Express the circuit as a canonical POS Boolean algebraic expression.
8. Express the canonical SOP expression in Σm notation.
9. Express the canonical POS expression in $\prod M$ notation.

10. Label the Boolean Algebraic postulate or theorems on the left with the step # from the right that best describes the transformation.

$$\text{Out} = \overline{A(\overline{A}B) + (\overline{A}B)B}$$

De Morgan _____

De Morgan _____

De Morgan _____

Distribution _____

Distribution _____

Inverse Axiom _____

Inverse Axiom _____

$$\text{Out} = \overline{A(\overline{A} + \overline{B}) + (\overline{A} + \overline{B})B} \quad (\#1)$$

$$= \overline{AA + AB + \overline{A}B + \overline{B}B} \quad (\#2)$$

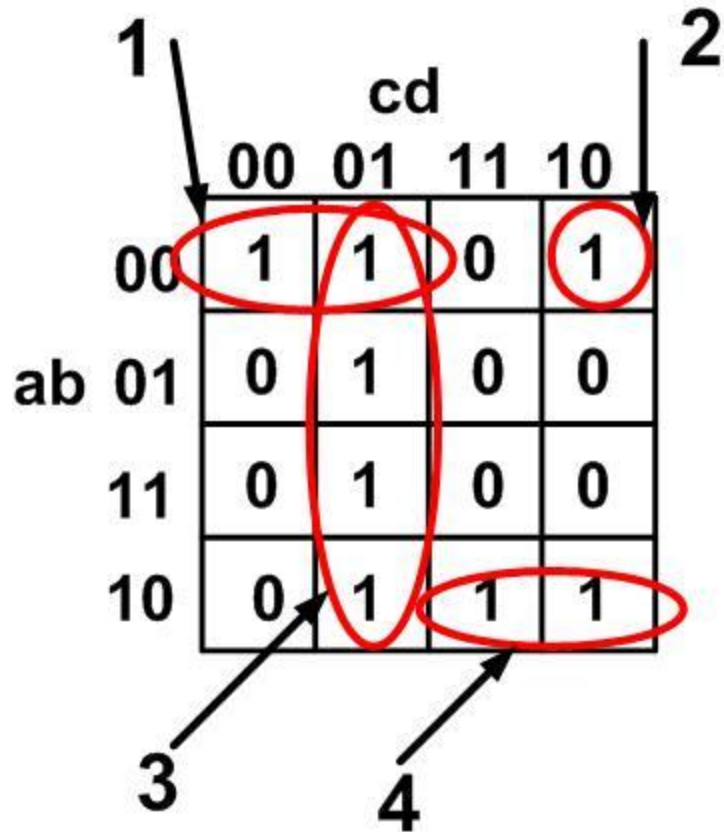
$$= \overline{AB + \overline{A}B} \quad (\#3)$$

$$= \overline{(AB)(\overline{A}B)} \quad (\#4)$$

$$= (\overline{A} + B)(A + \overline{B}) \quad (\#5)$$

$$= \overline{A}A + AB + \overline{A}\overline{B} + B\overline{B} \quad (\#6)$$

$$= AB + \overline{A}\overline{B} \quad (\#7)$$



11. Use the provided K-map. For each number corresponding to a section on the K-map, write all letters that apply.

1.

2.

3.

4.

- A- Is a minterm
- B- Is a implicant
- C- Is a prime implicant
- D- Is an essential prime implicant

12. Produce a hazard free Boolean algebraic expression for the following functions using K-maps. Begin with a minimized SOP expression. Then show which prime implicants get added to allow it to be static hazard free.

$$F(A,B,C) = \sum m(2,3,5,7)$$

$$F(A,B,C,D) = \sum m(1,2,3,4,5,6,9,10,11,12,13,14)$$

13. Given the following table of prime implicants and minterms, which of the prime implicants are essential? Why?

PI	0001	0011	0101	1010	1011	1100	1101	1110	1111
A'B'D	X	X							
A'C'D	X		X						
B'CD		X			X				
BC'D			X				X		
AC				X	X			X	X
AB						X	X	X	X

14. Given the table of prime implicants and minterms which of the following is true? Circle the correct statements.

1. You should remove dominating rows
2. You should remove dominated rows
3. You should remove dominating columns
4. You should remove dominated columns

Questions to ponder:

Why do you think we use base 10?

Pros and cons of Signed Magnitude vs. One's Complement vs. Two's Complement

There are 10 types of people in this world: those who understand binary and those that don't.

What are some cost metrics of for combinational circuits?